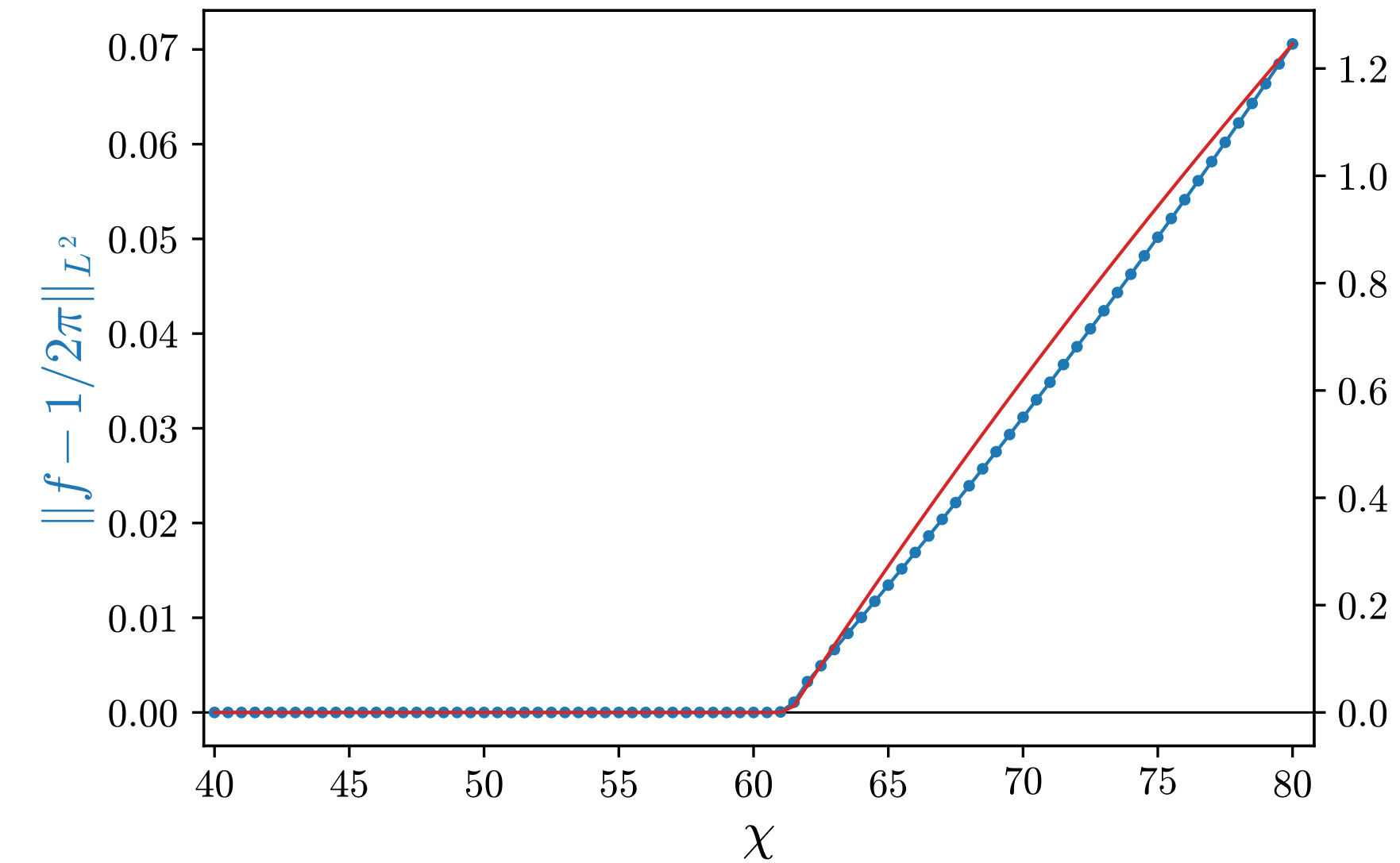


$$\begin{cases} \partial_t f = \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) = f_0 \end{cases}$$

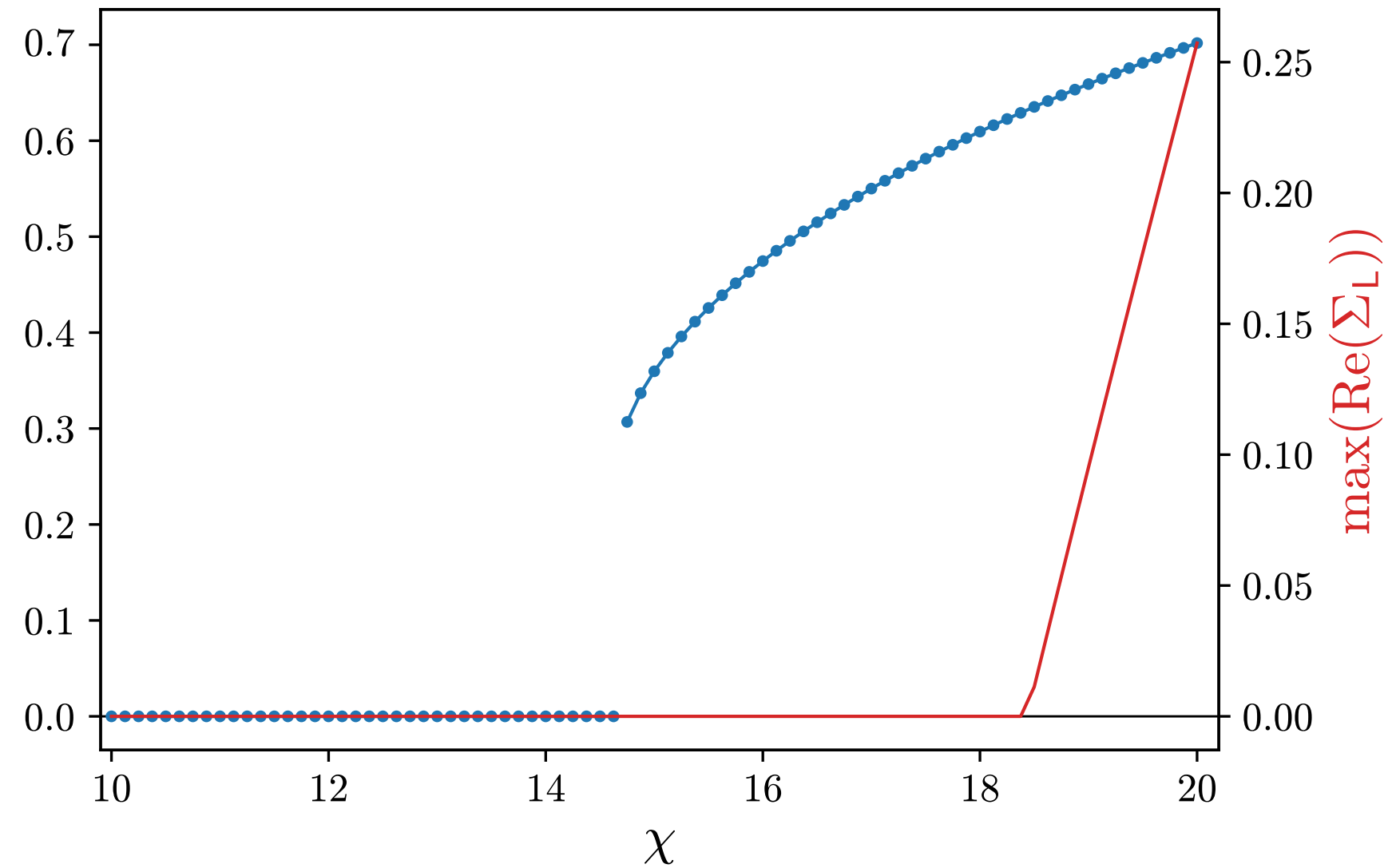
- What happens as $t \rightarrow \infty$?

• What is a super-critical transition?

Existence at or before loss of linear stability



$\tau < \tau^*$



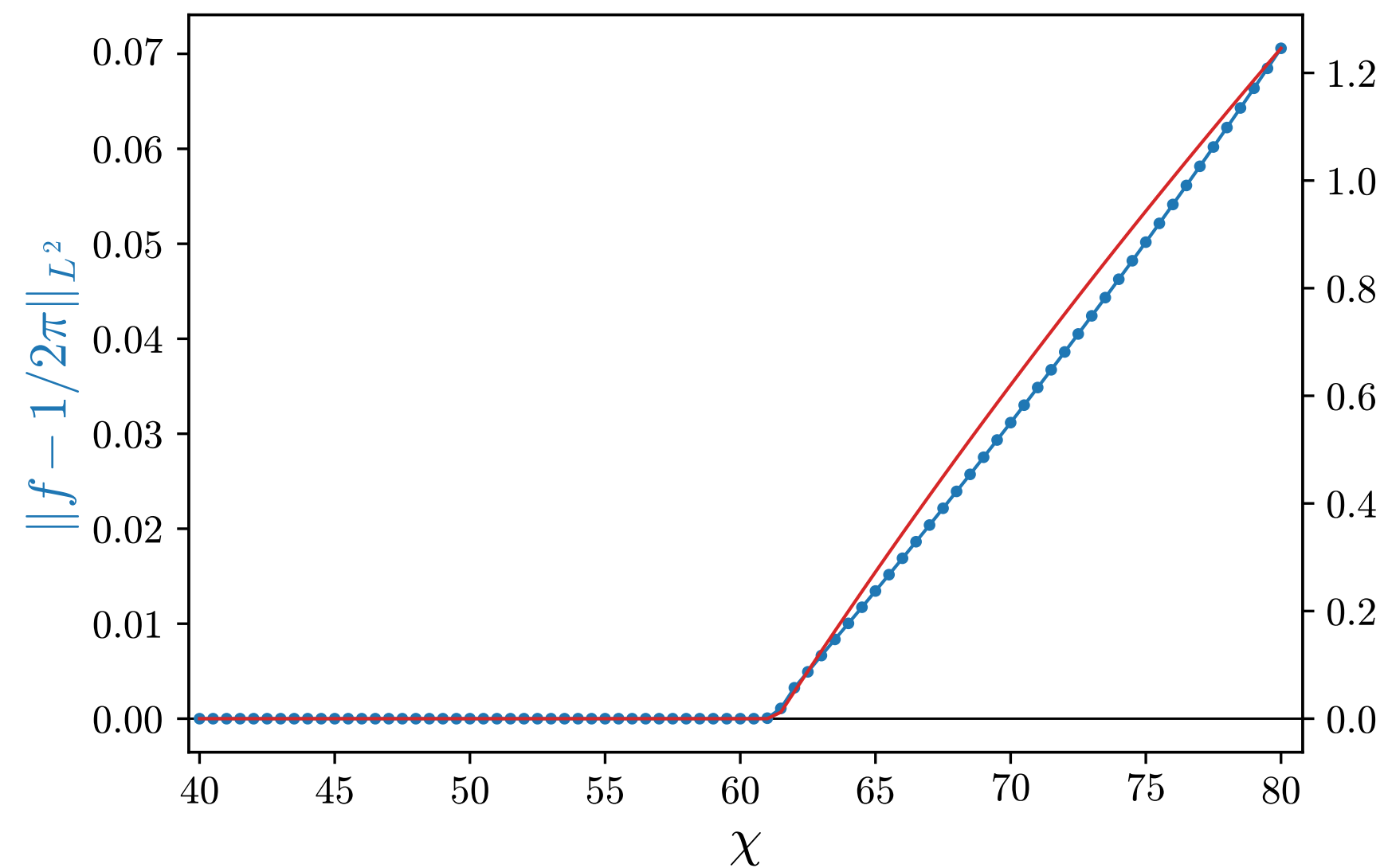
$\tau > \tau^*$

Dynamics

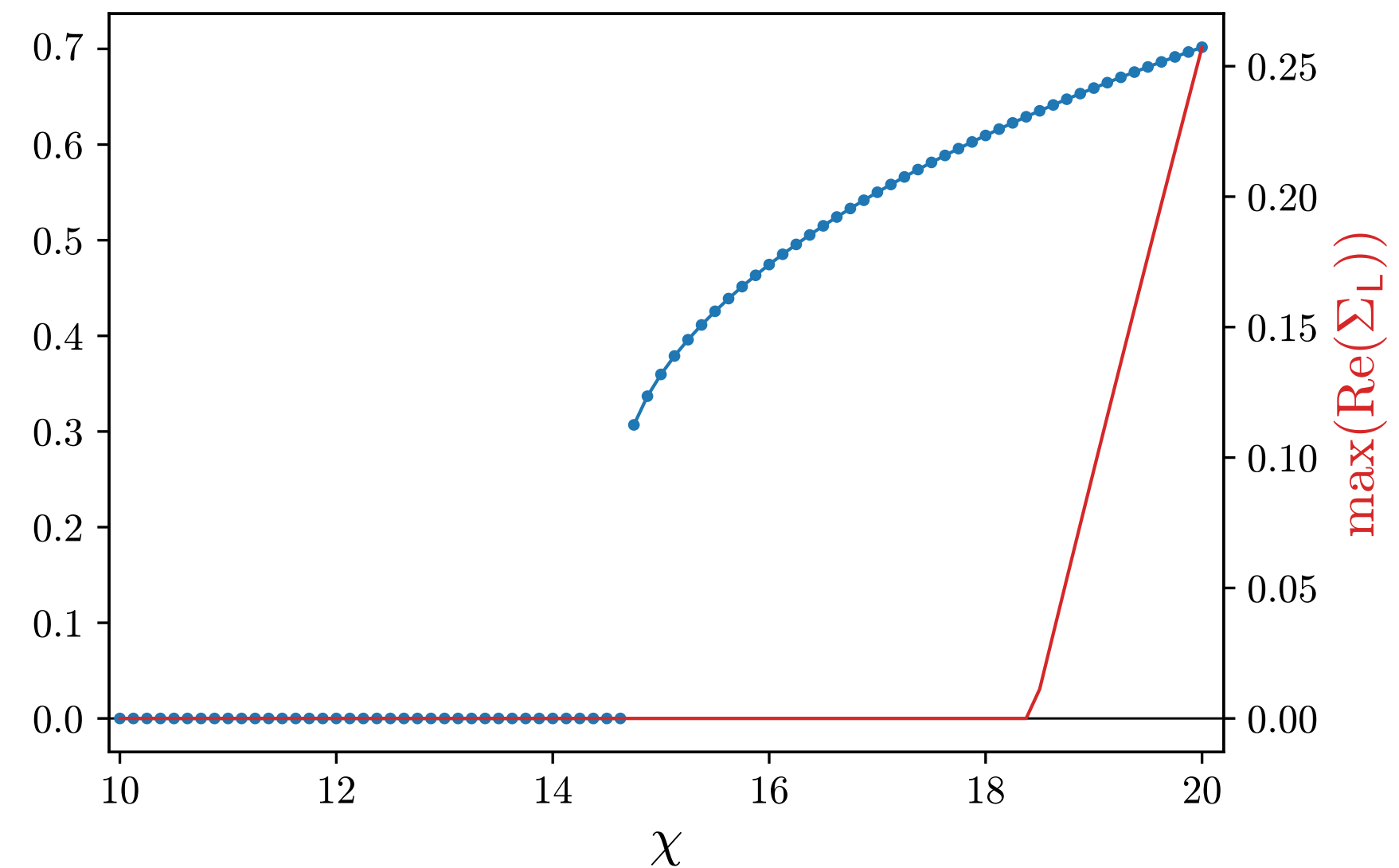
Dynamics

$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) &= f_0 \end{cases}$$

- What happens as $t \rightarrow \infty$?
- What is a super- to subcritical transition?
 - Existence *at or before* loss of local linear stability



$\tau < \tau^*$



$\tau > \tau^*$

Dynamics: non-uniform convergence?

$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta))) f \\ f(t=0) &= f_0 \end{cases}$$

$$\begin{cases} dX_t^i &= \text{Pe} v(\theta_t^i) dt + \sqrt{2\sigma_x} dW_t^i \\ d\theta_t^i &= \chi n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j) + \sqrt{2} dB_t^i \end{cases}$$

- Let $F_s = \{f_s(x, \theta)\}$ be translates of the spot solution and F_l the translates of the lane solution